CHAPTER
69

Pancreas

The pancreas is one of the largest digestive glands. The major part of the gland is exocrine, secreting enzymes involved in the digestion of lipids, carbohydrates and proteins. It has an additional endocrine function derived from clusters of cells scattered throughout the substance of the gland, which take part in glucose homeostasis and the control of upper gastrointestinal motility and function.

The healthy pancreas is creamy pink in colour, with a soft to firm consistency and lobulated surface. It is divided into a head, neck, body, tail and uncinate process on the basis of its anatomical relations (**Figs 69.1–69.3A**). In adults, the gland measures 12–15 cm in length and is shaped as a flattened 'tongue' of tissue lying in the retroperitoneum, thicker at its medial end (head) and thinner towards the lateral end (tail). The head lies within the 'C' loop of the duodenum and the remainder of the gland extends transversely and slightly cranially across the retroperitoneum and posterior to the stomach to the hilum of the spleen. It is 'draped' over the vertebral column and other retroperitoneal structures, forming a distinct shallow curve, with the neck and medial body lying most anteriorly. In adults, it has an average volume of 70–80 cm^3, although this varies considerably between individuals (with a range of 40–170 cm^3) and tends to be greater in males (Djuric-Stefanovic et al 2012). Pancreatic volume increases with age, to reach a peak in the fourth decade. From about 60 years of age, the gland atrophies and fatty connective tissue replaces exocrine tissue (Caglar et al 2012, Saisho et al 2007).

The ventral surface of the pancreas is covered by parietal peritoneum and is crossed by the root of the transverse mesocolon. A loose connective tissue layer immediately posterior to the pancreas, sometimes known the fusion fascia of Treitz in the region of the pancreatic head and the fusion fascia of Toldt in the region of the body and tail, contains vessels that supply the pancreas (Kimura 2000).

HEAD

The head of the pancreas lies to the right of the midline, anterior and to the right of the vertebral column, within the curve of the duodenum. It is the thickest and broadest part of the pancreas but is still flattened in the anteroposterior plane. Superiorly, it lies adjacent to the first part of the duodenum; close to the pylorus, however, where the duodenum is on a short mesentery, it overlaps the upper part of the head anteriorly. The duodenal border of the head is flattened, slightly concave and adherent to the second part of the duodenum, particularly around the duodenal papillae. The superior and inferior pancreaticoduodenal arteries lie adjacent to this region. The inferior border of the pancreatic head lies superior to the third part of the duodenum and is continuous with the uncinate process. The anterior surface of the head is covered by peritoneum and related to the origin of the transverse mesocolon (**Fig. 69.3B**). Posteriorly, the common bile duct is partially embedded within the head of the gland just proximal to where it joins the pancreatic duct near the major duodenal papilla (Burgard et al 1991, Nagai 2003). The posterior surface of the pancreatic head is also related to the inferior vena cava, the right crus of the diaphragm and the termination of the right gonadal vein (Ch. 76; see Fig. 76.5). Near the midline, the head of the pancreas becomes continuous with the neck.

NECK

The neck of the pancreas is approximately 2 cm wide and links the head and body. It is often the most anterior part of the gland and may be defined as that part of the pancreas lying anterior to the formation of the portal vein (usually the union of the superior mesenteric and splenic veins) in the transpyloric plane (**Fig. 69.4D,E**). This is a crucial relationship when evaluating pancreatic cancer because malignant involvement of these vessels may make resection impossible. The superior mesenteric vein and portal vein groove the posterior aspect of the neck. The inferior mesenteric vein joins the confluence of the superior mesenteric vein and splenic vein in one-third of individuals (Kimura 2000; see Fig. 69.7). The anterior surface of the pancreatic neck is covered by peritoneum and lies adjacent to the pylorus. The anterior superior pancreaticoduodenal branch of the gastroduodenal artery descends in front of the gland at the junction of the head and neck.

BODY

The body of the pancreas is the longest part of the gland and runs from the neck to the tail, becoming progressively thinner. It is slightly triangular in cross-section, and has anterior and posterior surfaces and superior and inferior borders.

Anterior surface The anterior surface is covered by peritoneum, which is reflected anteroinferiorly from the surface of the gland to be continuous with the posterior layer of the greater omentum and the transverse mesocolon (see Fig. 63.4). The two layers of the transverse mesocolon diverge along this surface (see Fig. 69.3B). Above the attachment of the transverse mesocolon, the anterior surface of the pancreas is separated from the stomach by the lesser sac. Inferiorly, it lies within the infracolic compartment, and its anterior relations include

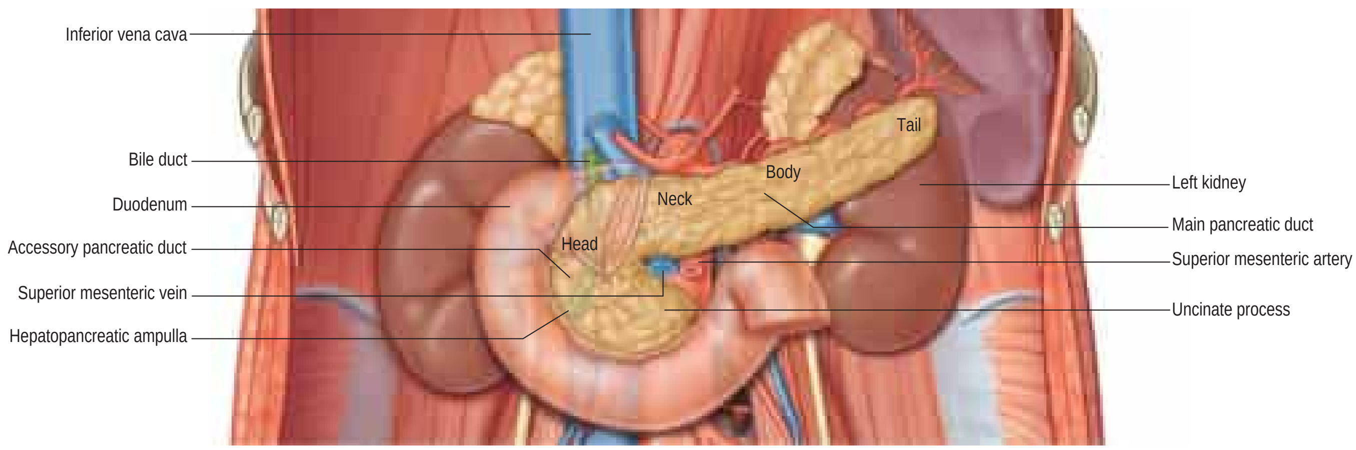


Fig. 69.1 Relations of the pancreas.

the fourth part of the duodenum, the duodenojejunal flexure and coils of jejunum.

Posterior surface The posterior surface of the pancreas is devoid of peritoneum. It lies on fascia (the fusion fascia of Toldt) anterior to the aorta and the origin of the superior mesenteric artery, the left crus of the diaphragm, left suprarenal gland, the upper pole of the left kidney surrounded by perirenal fascia, and left renal vein (Kimura 2000; see Figs 69.1, 69.4A–G). The splenic vein runs from left to right directly on this surface of the gland and indents the parenchyma to a variable extent, ranging from a shallow groove to almost a tunnel.

Superior border To the right, the superior border of the body of the pancreas is blunt but it becomes narrower and sharper to the left. An

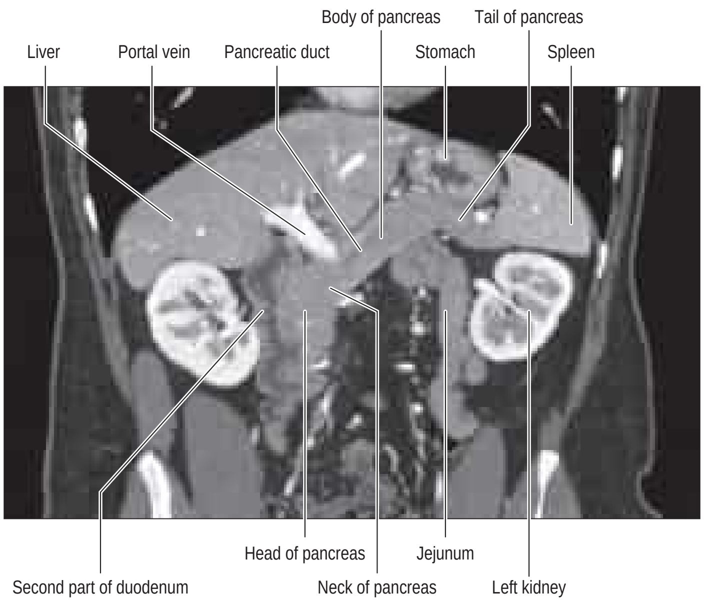


Fig. 69.2 A coronal reformat computed tomogram (CT) of the pancreas showing its relations in the upper abdomen. (Courtesy of the Department of Radiology, Global Hospital, Chennai, India.)

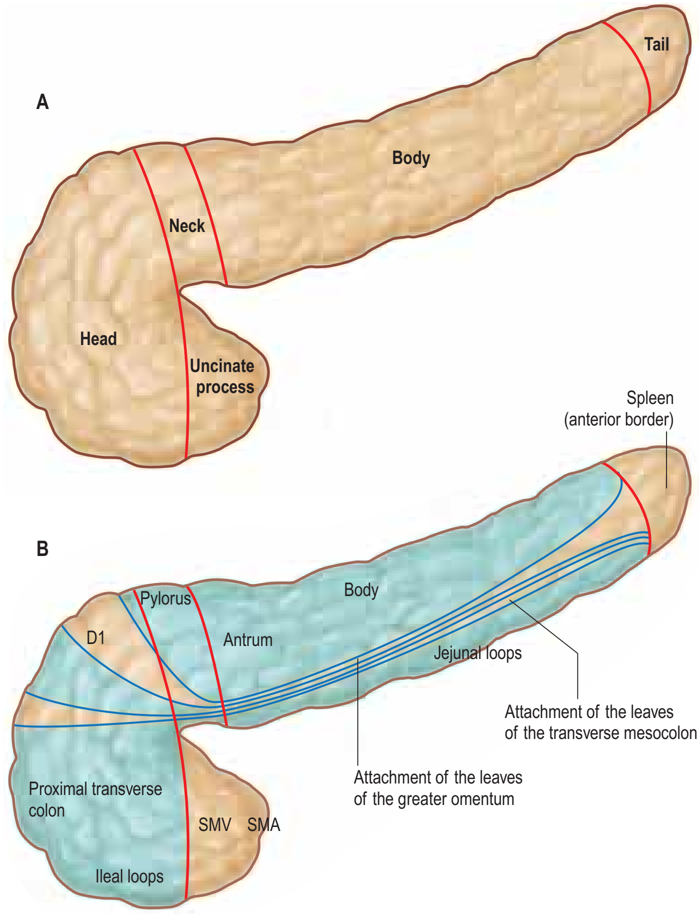
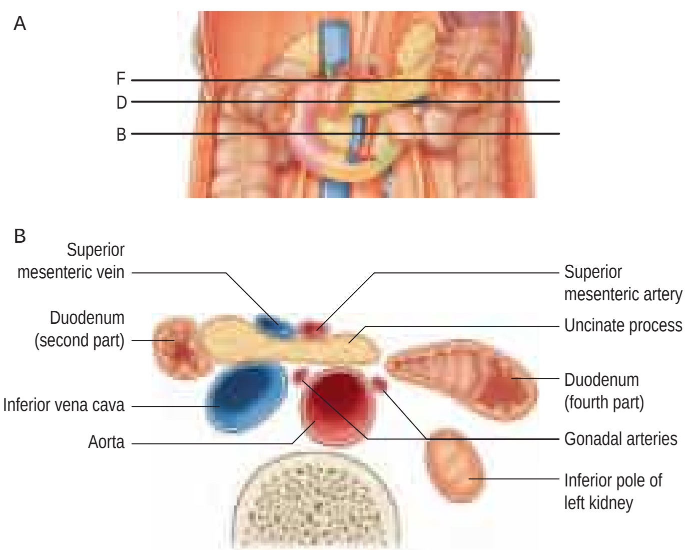


Fig. 69.3 **A**, Regions and anterior surfaces and borders of the pancreas. **B**, Anterior relations of the pancreas. Areas covered in peritoneum are shown in blue and structures overlying these areas are separated from the pancreas by peritoneal 'spaces'. The spleen lies anterior to the anterior leaf of the splenorenal ligament and not in direct contact with pancreatic tissue. Abbreviations: D1, first part of the duodenum; SMA, superior mesenteric artery; SMV, superior mesenteric vein.



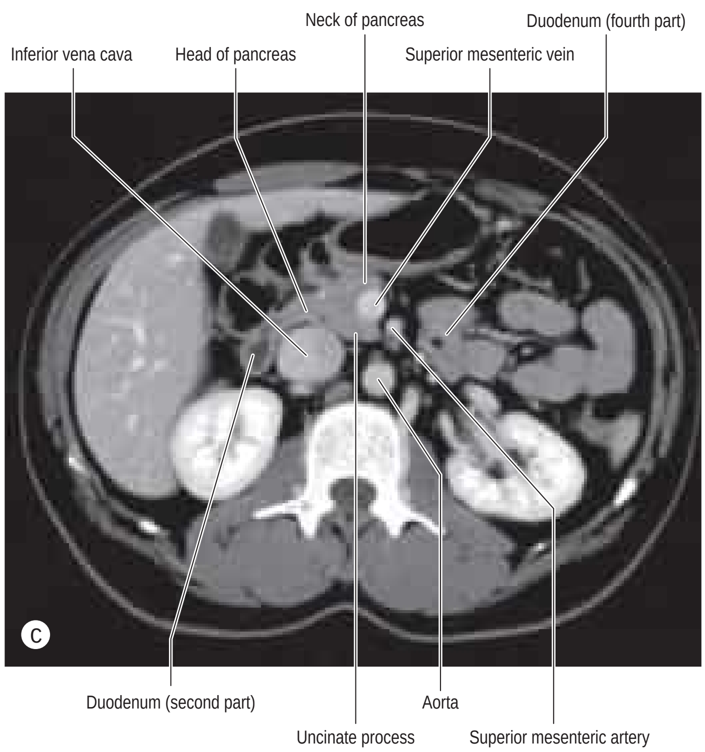


Fig. 69.4 **A**, Posterior relations of the pancreas, with planes of section labelled as shown in B, D and F. **B**, Diagrammatic cross-section taken at the mid level of the uncinate process, and equivalent CT axial slice (**C**).

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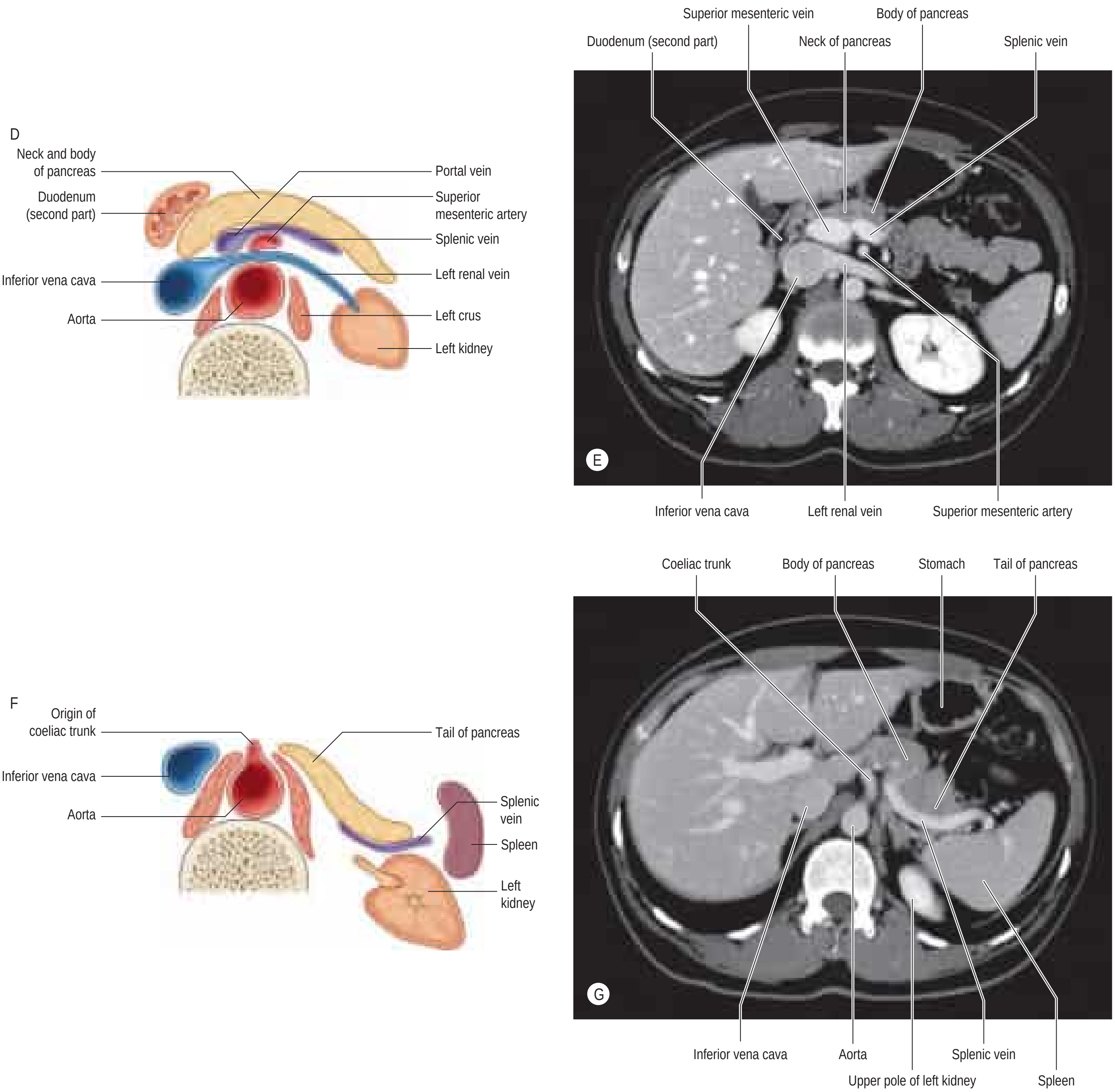

Fig. 69.4, cont'd Diagrammatic cross-sections and equivalent CT axial slices taken at the head and neck (**D–E**), and tail (**F–G**) of the pancreas.

omental tuberosity usually projects from the right end of the superior border above the level of the lesser curvature of the stomach. The superior border is related to the coeliac artery and its branches; the common hepatic artery runs to the right just above the gland, and the splenic artery passes to the left in a tortuous manner, projecting above the superior border at several points.

Inferior border At the medial end of the inferior border, adjacent to the neck of the pancreas, the superior mesenteric vessels emerge from behind the gland. Laterally, the inferior mesenteric vein runs behind the inferior border to join the splenic vein or the confluence of the splenic and superior mesenteric veins. This is a useful site for identification of the inferior mesenteric vein on computed tomographic (CT) imaging and during left-sided colonic resections.

TAIL

The tail of the pancreas is the narrowest, most lateral portion of the gland and is continuous medially with the body. It is between 1.5 and 3.5 cm long in adults and lies between the layers of the splenorenal ligament. It may terminate at the base of the splenorenal ligament or extend up to the splenic hilum, when it is at risk of injury at splenectomy during ligation or stapling of the splenic vessels. The splenic artery and its branches, and the splenic vein and its tributaries, lie posterior to the tail (see Fig. 69.4F,G).

UNCINATE PROCESS

The uncinate process is a hook-shaped continuation of the inferomedial part of the head of the gland. Embryologically, it is separate from the rest of the gland (p. 1051). The superior mesenteric vein and, occasionally, the superior mesenteric artery descend on its anterior surface before running forwards into the root of the mesentery of the small intestine. The uncinate process extends medially anterior to the abdominal aorta above the third part of the duodenum, which may be compressed by a pancreatic tumour at this site. On sagittal cross-sectional imaging, the left renal vein, uncinate process, and third part of duodenum can be seen lying between the superior mesenteric artery anteriorly and the abdominal aorta posteriorly.

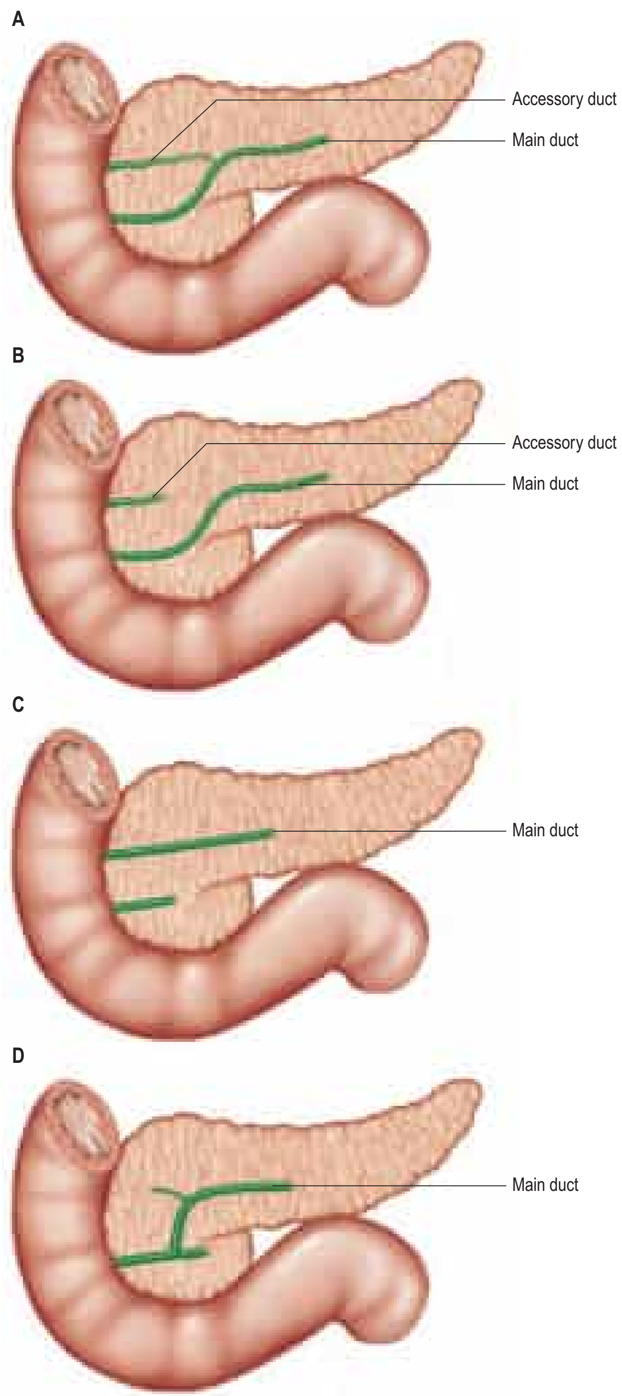

Fig. 69.5 Variations in the ductal anatomy of the pancreas. **A**, Normal. The main pancreatic duct is formed by fusion of the dorsal and ventral bud ducts, and communicates with the accessory pancreatic duct. **B**, Normal. There is no communication between the main duct and a normally sited accessory duct. **C**, Pancreas divisum. The majority of the pancreas drains via a dominant dorsal duct that enters the duodenum at the minor papilla. A short ventral duct is seen draining, along with the bile duct, into the major duodenal papilla. **D**, Absence of accessory duct. (E–G, *continued online*)

PANCREATIC DUCTS

The exocrine pancreatic tissue drains into multiple small lobular ducts. The arrangement of the main ducts draining the pancreas is subject to variation but the most common arrangement is a single main and a single accessory duct (**Fig. 69.5**). This arrangement reflects the embryological development of the dorsal and ventral pancreatic ducts (p. 1051; see Figs 60.3–60.4). The main pancreatic duct (of Wirsung) usually runs within the substance of the gland from left to right. It lies midway between the superior and inferior borders of the pancreas, usually slightly more towards the posterior surface of the gland. It is formed by the junction of several lobular (secondary) ducts in the tail and increases in calibre within the body of the gland as it receives further lobular ducts that join it almost at right angles to its axis, forming a 'herring-bone pattern'. In adults, the duct can often be demonstrated on ultrasound, measuring approximately 3 mm in diameter in the head, 2 mm in the body and 1 mm in the tail; the calibre of the duct increases from about the fifth decade onwards (Glaser et al 1987). As it reaches the neck of the gland, it turns inferiorly and posteriorly towards the bile duct, which lies on its right side. The two ducts unite to form a short common channel, which enters the wall of the descending part of the duodenum obliquely; it may contain a dilation known as the hepatopancreatic ampulla (of Vater) (p. 1175). The terminal part of the main pancreatic duct contains a few mucosal folds that impede reflux of pancreatic juice (Purvis et al 2013). The length of the common pancreaticobiliary channel is variable and measures up to 5–7 mm in normal individuals (Horaguchi et al 2014).

The accessory (dorsal) pancreatic duct (of Santorini) drains the upper part of the anterior portion of the pancreatic head. Much smaller in calibre than the main duct, it is formed within the substance of the head from several lobular ducts and usually communicates with the main pancreatic duct near the neck of the gland or near its first inferior branch (Kamisawa and Okamoto 2008, Kamisawa 2004). The accessory duct usually opens on to a small, rounded minor duodenal papilla, which lies about 2 cm proximal to the major papilla (see Fig. 69.5A). If the duodenal end of the accessory duct fails to develop, the lobular ducts drain via small channel(s) into the main duct (Hernandez-Jover et al 2011). The anatomy of the main and accessory pancreatic ducts may vary, reflecting anomalies in the development and fusion of the dorsal and ventral pancreatic ducts.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The pancreas has a rich arterial supply via branches from the coeliac trunk and superior mesenteric artery (**Figs 69.6A–C**; see also Fig. 65.4). The head and adjoining duodenum are supplied mainly by four arteries: two from the coeliac trunk via the gastroduodenal artery (anterior and posterior superior pancreaticoduodenal arteries), and two from the superior mesenteric artery via the inferior pancreaticoduodenal artery (anterior and posterior inferior pancreaticoduodenal arteries). The two anterior arteries supply the ventral aspects of the duodenum, pancreatic head and uncinate process, and join to form the anterior pancreaticoduodenal arcade. The two posterior arteries supply the dorsal aspect of the pancreatic head, adjacent duodenum and distal bile duct, and may join to form a posterior pancreaticoduodenal arcade.

The body and tail of the pancreas are supplied by multiple branches from the splenic artery, including the dorsal pancreatic artery. Small arteries characteristically run along the superior and inferior borders of the gland, lying in a deep groove or within the parenchyma. They are fed along their length by the pancreaticoduodenal and splenic arteries. Bleeding from these vessels is avoided by ligating the upper and lower borders of the pancreas prior to transection.

Posterior superior pancreaticoduodenal artery

The posterior superior pancreaticoduodenal artery usually arises as the first branch of the gastroduodenal artery at the superior edge of the first part of the duodenum. It runs to the right, anterior to the supraduodenal portion of the bile duct, before spiralling posteriorly around the common bile duct and descending on the posterior surface of the pancreatic head. It gives branches to the duodenum, head of the pancreas, and bile duct before anastomosing with the posterior inferior pancreaticoduodenal artery to form the posterior pancreaticoduodenal arcade. The posterior superior pancreaticoduodenal artery may also give origin to the retroduodenal and supraduodenal arteries, which supply the duodenum (Bertelli et al 1996a). The spiral course of the posterior superior pancreaticoduodenal artery reflects its embryonic development: it is the artery to the anterior aspect of the ventral pancreatic bud and the bile duct, which subsequently rotate clockwise behind the duodenum and dorsal pancreatic bud (Bertelli et al 1997).

Anterior superior pancreaticoduodenal artery

The anterior superior pancreaticoduodenal artery usually arises as the smaller terminal branch of the gastroduodenal artery, together with the right gastroepiploic artery. It originates behind the first part of the duodenum and runs inferiorly on the anterior aspect of the pancreatic head at a variable distance medial to the groove between the descending duodenum and pancreatic head. It passes inferiorly around the right or inferior border of the pancreatic head, often piercing the gland to reach the posterior surface of the uncinate process, where it joins the anterior

An abnormally long common channel may occur in isolation or with congenital choledochal dilation, and is associated with pancreaticobiliary reflux and an increased risk of gallbladder cancer (Kamisawa et al 2010a).

Patients who develop acute pancreatitis are more likely to have a poorly developed minor papilla; if this were not so, the patent accessory duct system might act as a secondary drainage mechanism for the main ductal system and so prevent elevation of intraductal pressures likely to cause acute pancreatitis (Kamisawa et al 2010b). One clinically important variation is 'pancreas divisum', seen in 5–10% of individuals (Rana et al 2012; see Fig. 69.5C, G). In this condition, there is a failure of normal duct fusion and the accessory pancreatic duct drains the entire pancreas via the minor papilla, except that part of the head and uncinate process that develop from the ventral bud. The relatively small size of the minor papilla might impair pancreatic drainage, leading to recurrent pancreatitis.

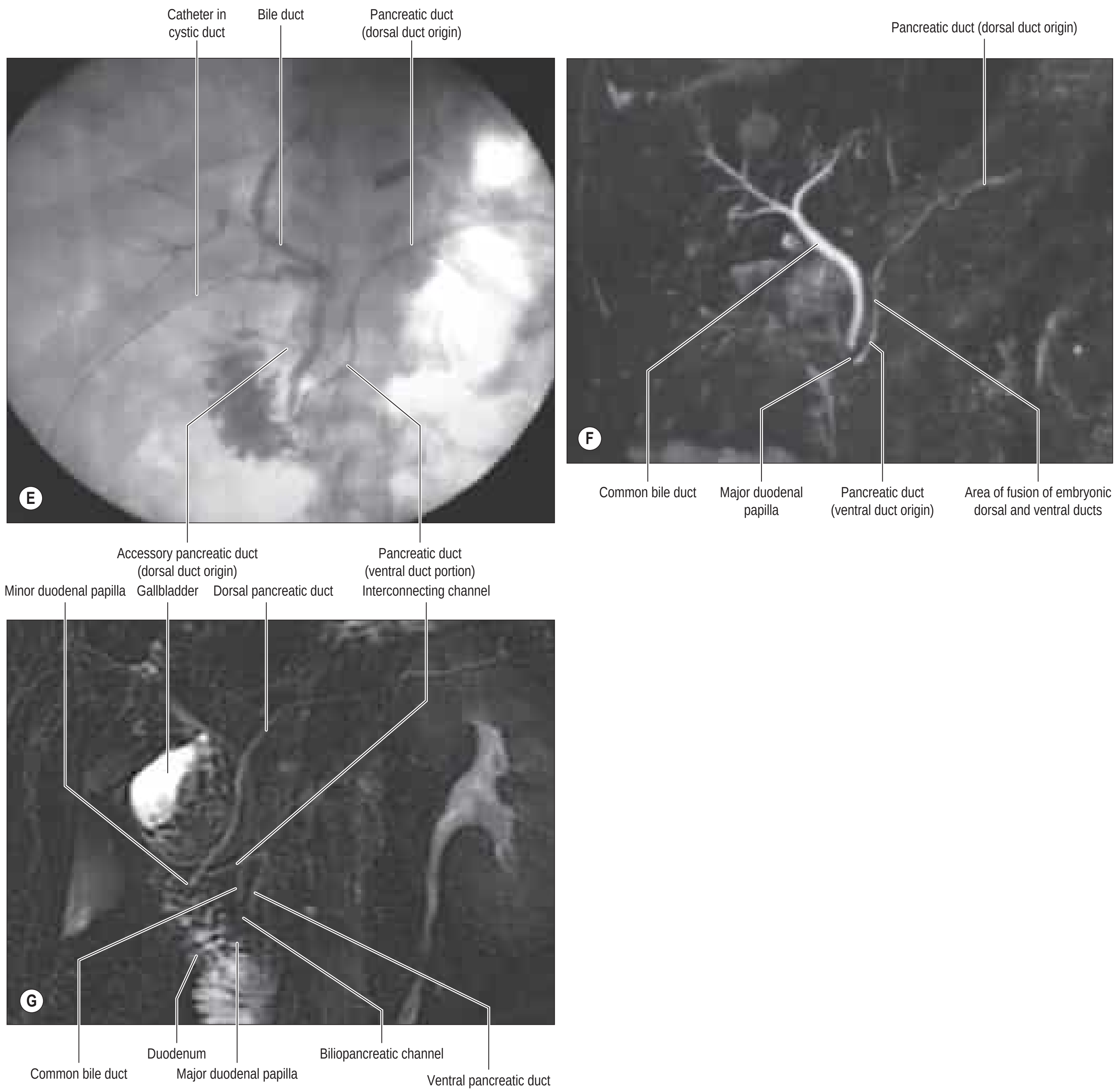

Fig. 69.5, cont'd Variations in the ductal anatomy of the pancreas. **E**, Intraoperative cholangiogram visualizing the main pancreatic duct opening into the duodenum through the major duodenal papilla, the accessory duct opening through the minor duodenal papilla and its communicating branch. **F**, Normal magnetic resonance cholangiopancreatogram (MRCP). The dorsal duct joins with the distal ventral duct in the head of the pancreas. **G**, Pancreas divisum. MRCP reconstruction shows that the dorsal duct is dominant and enters the duodenum at the minor duodenal papilla. A short ventral pancreatic duct is seen draining with the bile duct into the major papilla. (Figs 69.5E–G, courtesy of Mohamed Rela, Global Hospitals, Chennai, India.)

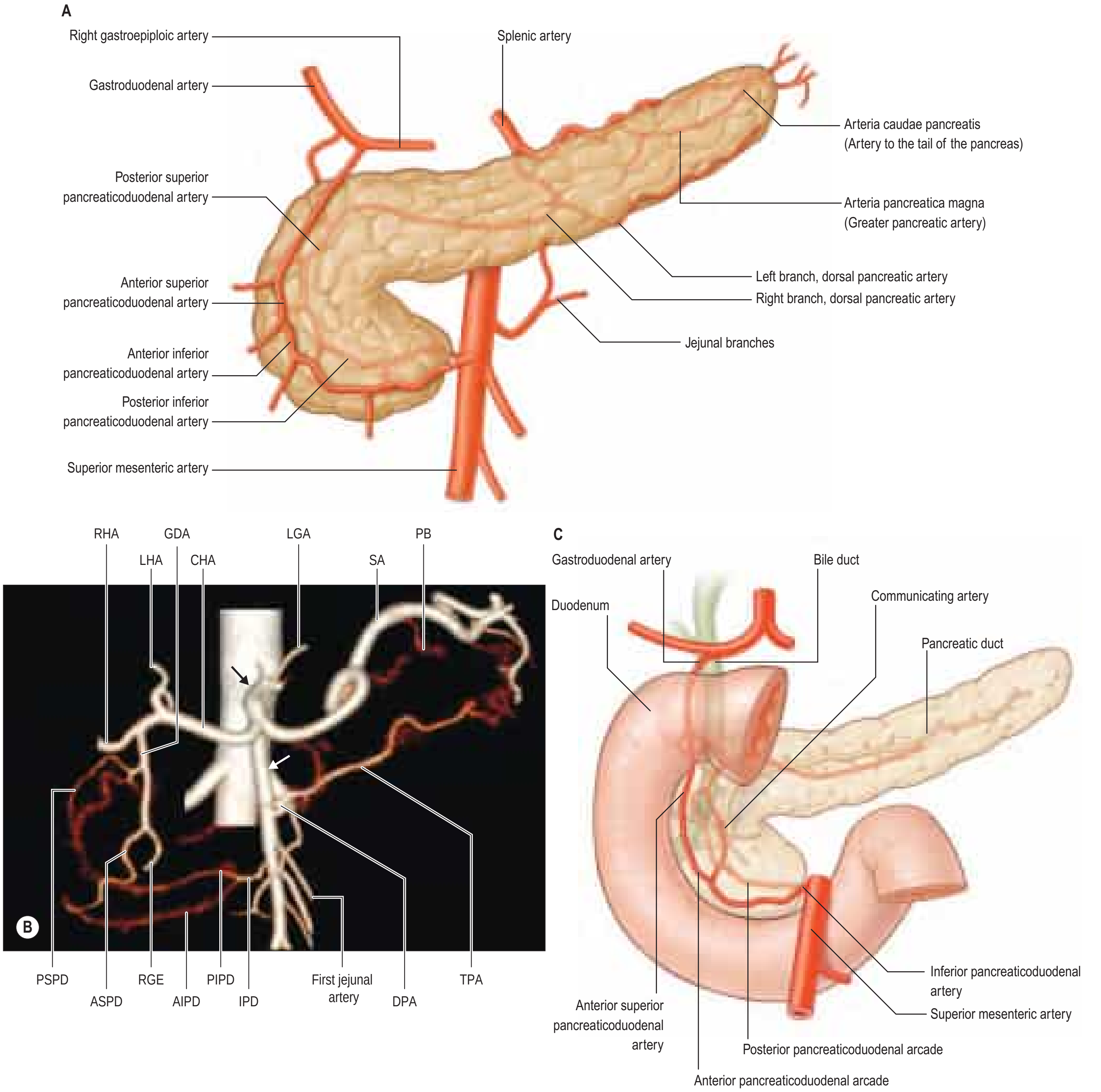

Fig. 69.6 **A**, The arterial supply of the pancreas. **B**, A CT angiographic reconstruction. The black arrow shows the coeliac trunk, and the white arrow shows the superior mesenteric artery. Anastomotic arcades, both ventral and dorsal, can be seen coursing around the head of the pancreas. The body and tail of the pancreas are supplied by multiple pancreatic branches (PB) from the splenic artery and the dorsal pancreatic artery (DPA), which, in this example, arises from the superior mesenteric artery, and gives origin to the transverse pancreatic artery (TPA). Other abbreviations: AIPD, anterior inferior pancreaticoduodenal artery; ASPD, anterior superior pancreaticoduodenal artery; CHA, common hepatic artery; GDA, gastroduodenal artery; IPD, inferior pancreaticoduodenal artery; LGA, left gastric artery; LHA, left hepatic artery; PIPD, posterior inferior pancreaticoduodenal artery; PSPD, posterior superior pancreaticoduodenal artery; RGE, right gastroepiploic artery; RHA, right hepatic artery; SA, splenic artery. **C**, The anterior and posterior pancreaticoduodenal arterial arcades and a typical communicating artery. (Adapted with permission from Mirjalili SA, Stringer MD. The arterial supply of the major duodenal papilla and its relevance to endoscopic sphincterotomy. Endoscopy 2011; 43(4): 307–11.)

inferior pancreaticoduodenal artery to form the anterior pancreaticoduodenal arcade (Bertelli et al 1995).

Inferior pancreaticoduodenal artery

An inferior pancreaticoduodenal artery is present in most individuals. It usually arises either directly from the superior mesenteric artery at the inferior border of the pancreas or as a common vessel with the first jejunal artery (a pancreaticoduodenojejunal trunk) from the posterior or left aspect of the superior mesenteric artery (Horiguchi et al 2008, Bertelli et al 1996b). It runs to the right, posterior to the superior mesenteric vein, to reach the posterior surface of the uncinate process, where it divides into anterior and posterior inferior pancreaticoduodenal arteries. When arising as a pancreaticoduodenojejunal trunk, the artery gives off a jejunal branch and then runs posterior to both the superior mesenteric artery and vein before dividing into its terminal branches. Occasionally, the inferior pancreaticoduodenal artery is absent and the anterior and posterior inferior pancreaticoduodenal arteries arise separately from the superior mesenteric artery.

The smaller anterior inferior pancreaticoduodenal artery runs to the right on the posterior surface of the uncinate process to join the anterior superior pancreaticoduodenal artery and form the anterior pancreaticoduodenal arcade. The larger posterior inferior pancreaticoduodenal artery runs on the posterior surface of the head of the pancreas, parallel and superior to the anterior inferior pancreaticoduodenal artery. It joins the posterior superior pancreaticoduodenal artery to form the posterior pancreaticoduodenal arcade (Bertelli et al 1997).

CHAPTER 69

Communicating arteries

Multiple small arteries run between the anterior and posterior pancreaticoduodenal arcades, either via the pancreaticoduodenal groove or through the substance of the gland. The largest and most consistent of these is the communicating artery (sometimes known as the middle pancreaticoduodenal arcade), which passes between the main and accessory pancreatic ducts and connects the anterior pancreaticoduodenal arterial arcade and the posterior superior pancreaticoduodenal artery (Yamaguchi et al 2001; **Fig. 69.6C**; see also Fig. 65.4). This artery gives rise to the majority of small arteries that supply the major duodenal papilla (Mirjalili and Stringer 2011).

Other arterial arcades in the head of the pancreas

Kirk's arcade is formed by a right branch of the dorsal pancreatic artery that emerges on to the anterior surface of the head of the gland and anastomoses with a branch of either the gastroduodenal or the anterior pancreaticoduodenal arcade (Woodburne and Olsen 1951). It supplies blood to the ventral surface of the head and neck of the gland.

Splenic artery

The splenic artery arises from the coeliac trunk and runs behind the superior border of the pancreas to the splenic hilum (see Fig. 69.6A). It gives multiple small branches along its course that penetrate and supply the pancreatic parenchyma. Prominent among these are the dorsal pancreatic artery (see below); great pancreatic artery (arteria pancreatica magna), arising approximately two-thirds of the way along the gland; and the artery to the tail of the pancreas (arteria caudae pancreatis), arising near the tail. These branches lie on or within the posterior aspect of the gland and often anastomose with the transverse pancreatic artery. Anatomical variations are not unusual; the dorsal, great or transverse pancreatic arteries may be dominant in any one individual (Wu et al 2011).

Dorsal pancreatic artery

The dorsal pancreatic artery commonly arises from the initial 2 cm of the splenic artery, although it may take origin from the common hepatic or superior mesenteric artery or the coeliac trunk (Horiguchi et al 2008). The artery is short and gives off numerous branches, including a terminal left branch near the inferior border of the gland. Several right-sided branches run to the head of the pancreas, passing either behind or in front of the superior mesenteric vein to supply the posterior or anterior surface of the pancreatic head, respectively; they anastomose with arteries of the pancreaticoduodenal arcade (Tsutsumi et al 2014). The terminal left branch anastomoses with the transverse pancreatic artery.

The length and course of the dorsal pancreatic artery depend primarily on its site of origin. When it arises from the splenic artery, common hepatic artery or coeliac trunk, the artery runs inferiorly on the dorsal surface of the pancreas, whereas if it arises from the superior mesenteric artery, it runs superiorly. The artery uniformly terminates near the inferior border of the pancreas close to the confluence of the splenic and superior mesenteric veins (Bertelli et al 1998).

Transverse pancreatic artery

The transverse pancreatic artery, also called the inferior pancreatic artery, commonly originates from the left terminal branch of the dorsal pancreatic artery. It runs to the left on the posterior surface of the gland close to its inferior border, gives multiple branches to the body and tail, and anastomoses with other pancreatic branches of the splenic artery.

The transverse pancreatic artery may occasionally originate from the gastroduodenal artery or the anterior superior pancreaticoduodenal artery, crossing the anterior surface of the pancreatic head to reach the inferior border of the neck of the gland and then the tail. In such cases, it may be the dominant artery to the pancreatic body, and its injury or deliberate ligation during pancreaticoduodenectomy may cause ischaemia of the remnant pancreas.

Arterial segmentation of the pancreas

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Venous drainage of the pancreas

Multiple veins drain the pancreas into the portal, superior mesenteric and splenic veins (Mourad et al 1994; **Fig. 69.7**). Although variations are common, the anterior superior pancreaticoduodenal vein typically joins either the confluence of the right gastroepiploic and right colic veins (the gastrocolic trunk of Henle) or the right gastroepiploic vein alone, to drain into the superior mesenteric vein at the inferior border of the neck of the pancreas (Mourad et al 1994, Lange et al 2000). The posterior superior pancreaticoduodenal vein drains cranially into the portal vein. Both anterior and posterior inferior pancreaticoduodenal veins usually drain directly into the superior mesenteric vein. Veins from the body and tail of the gland drain directly into the splenic vein. A transverse (or inferior) pancreatic vein may be present, running along the inferior border of the pancreas to join the inferior mesenteric vein. A few pancreatic veins communicate with systemic veins in the retroperitoneum (veins of Retzius); these may form retroperitoneal varices in portal hypertension.

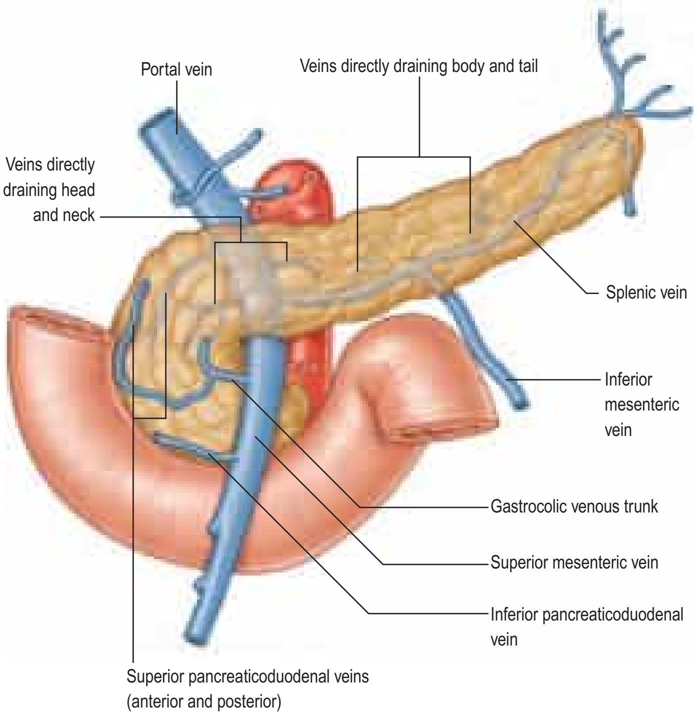

Fig. 69.7 Venous drainage of the pancreas. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Lymphatic drainage

The lymphatic drainage of the pancreas is extensive (see Fig. 64.14B), which, in part, explains the poor prognosis of pancreatic cancer. Lymphatic vessels commence within the connective tissue septa within the gland and join to form larger lymphatics that follow local arteries (Deki and Sato 1988, O'Morchoe 1997). Lymphatics from the tail and body drain mostly into nodes along the splenic artery and inferior border of the gland, and from there to pre-aortic nodes between the coeliac trunk and superior mesenteric artery. Lymphatics from the neck and head drain more widely into nodes along the pancreaticoduodenal, superior mesenteric and hepatic arteries, and, ultimately, into pre-aortic nodes (Donatini and Hidden 1992). There is no evidence of lymphatic channels within the pancreatic islets.

INNERVATION OF THE PANCREAS

The pancreas has a rich autonomic nerve supply (Love et al 2007). Parasympathetic afferents convey sensory information from pancreatic ducts, acini and islets via the vagus nerve. Preganglionic vagal efferents have their cell bodies in the dorsal motor nucleus of the vagus and reach the pancreas via hepatic, gastric and coeliac branches of the nerve. They synapse with postganglionic parasympathetic pancreatic neurones, which are distributed singly or clustered in ganglia in the interlobular connective tissue, lobules and islets. Pancreatic ganglia are most abundant in the head and neck of the gland. The ganglia contain neurones that are dominantly cholinergic but nitrergic, peptidergic and dopaminergic neurones are also present. These pancreatic neurones receive input not only from vagal efferents but also from enteric neurones from the stomach and duodenum, sympathetic efferents and other pancreatic neurones. Their nerve fibres ramify in the interacinar spaces and supply acinar cells and islets, modulating both exocrine and endocrine secretion (see Fig. 69.10). A network of

The pancreas may be divided into two arterial ‘segments’ that are separated by a relatively avascular plane; there is disagreement on the definition of these segments. One view is that the two segments are separated by a plane passing to the left of the superior mesenteric artery, such that the head and neck are supplied by branches of the gastroduodenal and superior mesenteric arteries, and the body and tail are supplied by branches of the splenic artery (Busnardo et al 1988). An alternative view holds that the anterior segment consists of the superior part of the pancreatic head, neck, body and tail of the gland (derived from the dorsal pancreatic bud), and the posterior segment consists of the inferior part of the head and the uncinate process (derived from the ventral pancreatic bud) (Sakamoto et al 2000). In both schemes, the main pancreatic duct crosses between the segments. Currently, segmental anatomy is of limited practical surgical significance.

interstitial cells (of Cajal) provides an additional link between autonomic nerves and pancreatic acini.

Visceral afferents transmit pain and other sensory information to cell bodies in the sixth to twelfth thoracic dorsal root ganglia via the coeliac plexus and thoracic splanchnic nerves. The cell bodies of preganglionic sympathetic nerves are located in the intermediolateral columns of the spinal cord (T6–12); their axons travel in the thoracic splanchnic nerves and synapse in the coeliac and superior mesenteric ganglia. Postganglionic sympathetic nerves innervate the blood vessels, ganglia and ducts within the pancreas, causing vasoconstriction and inhibiting exocrine secretion.

Perineural invasion is relatively common in pancreatic cancer and not only adversely affects prognosis but may also be a factor in the pain associated with this disease (Bapat et al 2011).

Referred pain

Pain arising in the pancreas is poorly localized. In common with other foregut structures, the majority of pain arising from the pancreas is referred to the epigastrium. Inflammatory or infiltrative processes arising from the gland rapidly involve the tissues of the retroperitoneum that are innervated by somatic nerves; pain is then localized to the lower thoracic spine. Intractable pain from chronic pancreatitis or inoperable pancreatic tumours can be temporarily controlled by thermal or chemical ablation of the coeliac plexus.

MICROSTRUCTURE

The pancreas is composed of exocrine and endocrine tissues. The main tissue mass is exocrine, forming 98% of the pancreas, in which pancreatic islets, forming the endocrine component, are embedded (Figs 69.8–69.10).

Exocrine pancreas

The exocrine pancreas is a branched acinar gland (see Fig. 2.6), surrounded and incompletely lobulated by delicate loose connective tissue. Individual acini are composed of roughly spherical clusters of pyramidal cells that secrete digestive enzymes, water and bicarbonate. A narrow intralobular duct lined by flattened or cuboidal centro-acinar cells originates from within each secretory acinus. Intralobular ducts unite to form larger interlobular ducts lined by taller cuboidal and, eventually, columnar epithelium. The latter are surrounded by connective tissue septa, containing smooth muscle, autonomic nerves and fat. Neuroendocrine cells are present among the columnar ductal cells, and mast cells are numerous in the surrounding connective tissue. A network of intralobular arteries, arterioles and capillaries form capillary networks around the terminal ducts, acini and the richly vascularized endocrine islets (Love et al 2007).

Acinar cells Acinar cells have a basal nucleus surrounded by abundant basophilic rough endoplasmic reticulum and an apical region containing dense eosinophilic secretory zymogen granules. A prominent supranuclear Golgi complex is associated with large, membrane-bound granules containing the protein constituents of pancreatic secretion, including enzymes that are only active after release. Ganglionic neurones and cords of undifferentiated epithelial cells are also found within the acini. The structure of the exocrine pancreas and its functional regulation are summarized in Figure 69.8.

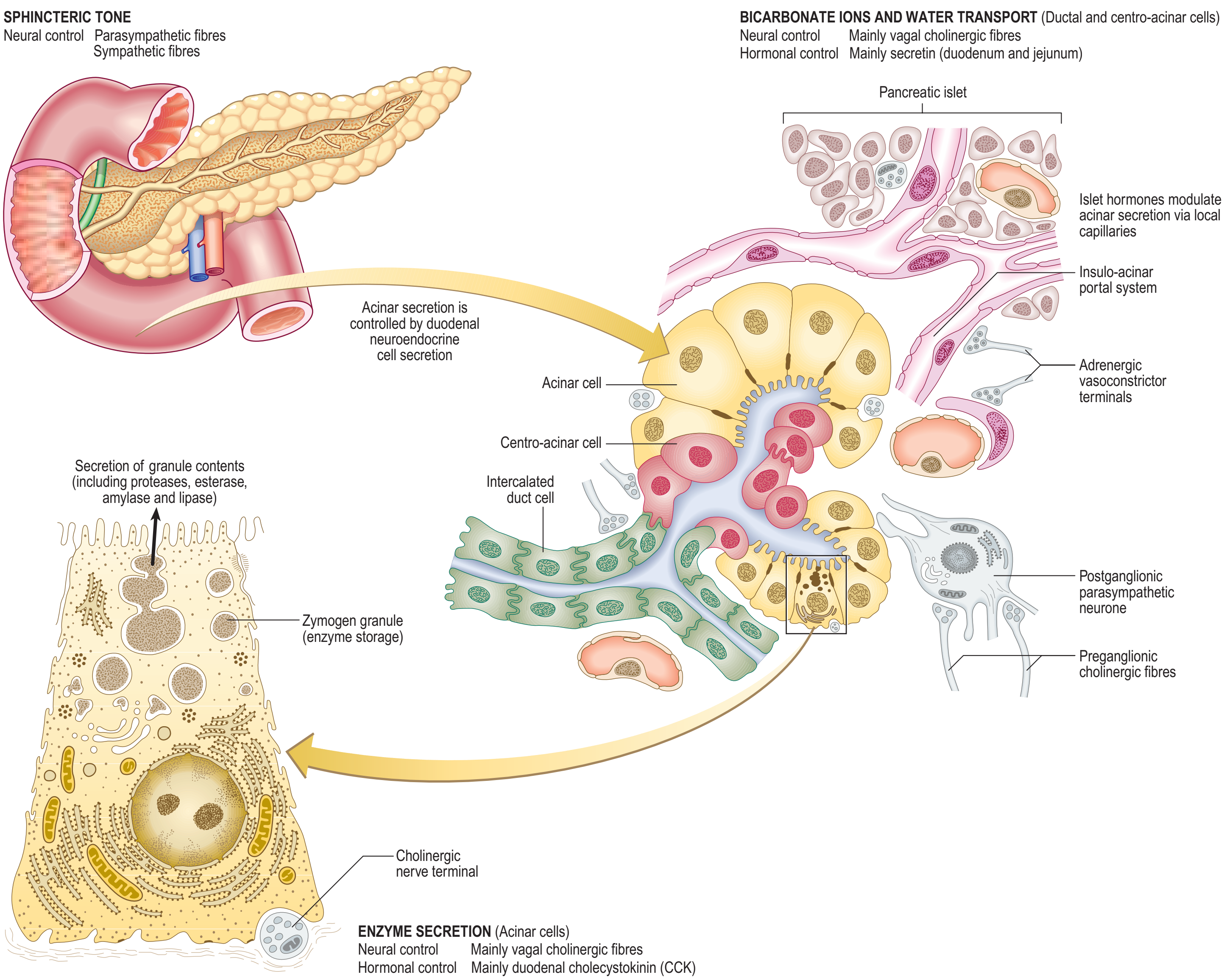

Fig. 69.8 The microstructure of the exocrine pancreas and the mechanisms by which its secretion is controlled. Pancreatic stellate cells (see text) are not shown.

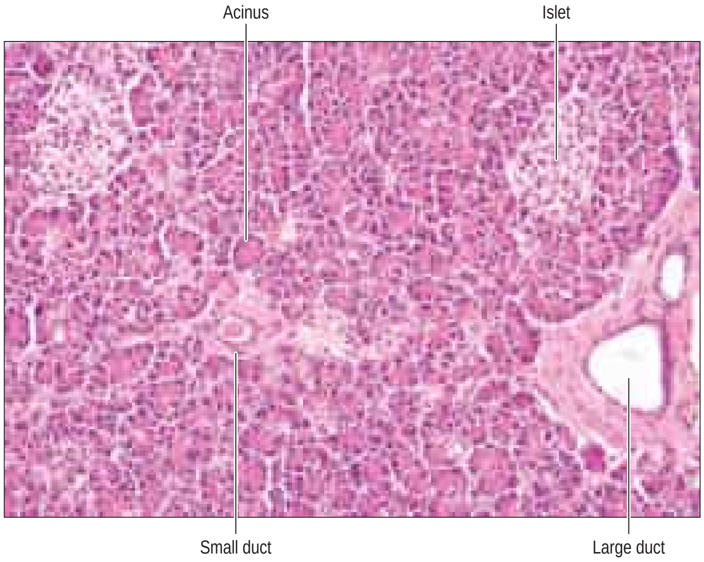

Fig. 69.9 Microstructure of the pancreas. The exocrine tissue consists of acini with pyramid-shaped columnar cells arranged in spherical clusters. Acinar secretions drain into intralobular ducts, which, in turn, join larger interlobular ducts, present in connective tissue septa containing blood vessels. Interspersed between the acini are islets of Langerhans, which appear as clusters of pale-staining cells surrounded by a network of capillaries, seen as narrow clear spaces. Adipocytes may also be present. Haematoxylin and eosin. (Courtesy of Dr Mukul Vij, Global Hospital, Chennai, India.)

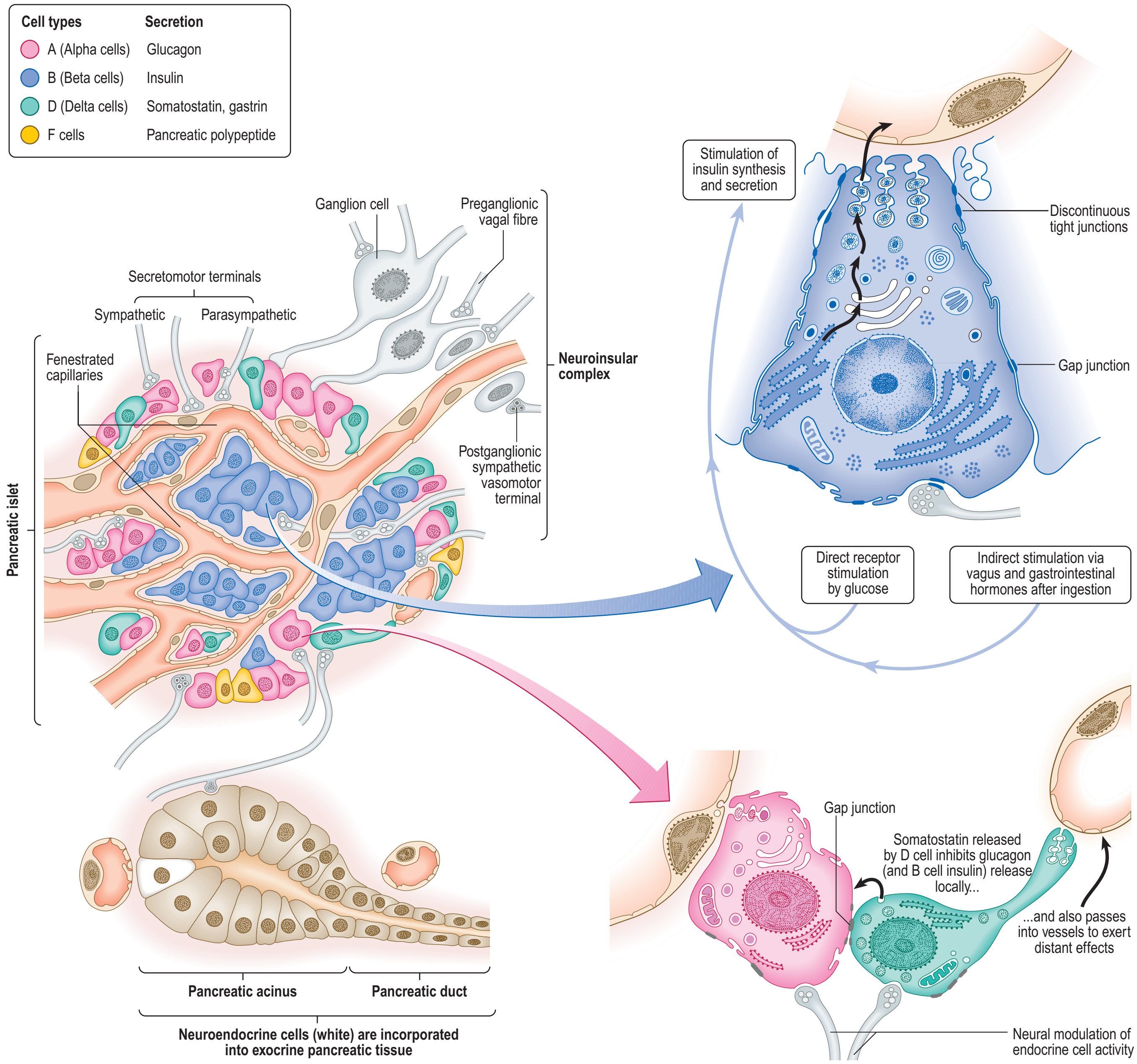

Fig. 69.10 The microstructure and control of function of the endocrine pancreas.

Stellate cells Pancreatic stellate cells are myofibroblast-like cells distributed in the exocrine part of the pancreas, mainly in the peri-acinar space, where their long cytoplasmic processes encircle the base of the acinus, and in perivascular and periductal regions of the pancreas. They are usually in a quiescent state but, when activated, they have been implicated in the pathogenesis of chronic pancreatitis and pancreatic cancer (Omary et al 2007, Masamune and Shimosegawa 2013).

Endocrine pancreas

Pancreatic islets (of Langerhans) constitute the endocrine component of the pancreas. The human pancreas may contain more than a million islets distributed throughout the gland but most numerous in the tail. Each islet consists of a highly vascularized cluster of polyhedral cells arranged in a trabecular pattern, closely approximated to a dense network of fenestrated capillaries, which enables bidirectional transport of substrates between the islet cells and the circulation (In't Veld and Marichal 2010). Islets are also supplied by autonomic nerve fibres, which travel with the blood vessels and end blindly in the pericapillary space. True synaptic contact between these nerve fibres and islet cells has not been demonstrated, though their role in islet function has been suggested (Rodriguez-Diaz et al 2012). Specialized staining techniques are necessary to distinguish various islet cells, designated alpha, beta, delta, epsilon and pancreatic polypeptide (PP) or F cells. Beta cells secrete insulin and are the most abundant (nearly 50% in human islets), followed by glucagon-secreting alpha cells (30–40%). Delta cells and PP cells are less common and secrete somatostatin and pancreatic polypeptide, respectively. Epsilon cells secrete ghrelin. Unlike mouse islets, where beta cells form the core of the islet and are surrounded by alpha cells, human islets do not show cell-type specific localization. The various cell types are uniformly distributed along the microvasculature and have extensive interlinkages with other islet cells. Their general organization is shown in Figure 69.10.

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Fig. 69.5 E–G Variations in the ductal anatomy of the pancreas.

Fig. 69.11 An intraoperative photograph taken during pancreaticoduodenectomy.

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Acute inflammation of the pancreas occurs in response to a variety of pancreatic insults, including gallstone obstruction of the hepatopancreatic ampulla, which causes reflux of bile and pancreatic juice into the pancreatic duct. This precipitates intraparenchymal activation of inactive digestive enzymes, leading to cell injury, an inflammatory cascade and destruction of pancreatic tissue.

The regional anatomy of the pancreas explains most of the clinical complications of acute pancreatitis. Irritation of the stomach contributes to gastric stasis and vomiting; inflammation of the superior mesenteric plexus may contribute to paralytic ileus; and vascular inflammation may cause superior mesenteric or portal vein thrombosis, arterial bleeding or pseudoaneurysms, and ischaemia of the transverse colon. Haemorrhagic fluid may accumulate in the loose retroperitoneal tissues around the pancreas and track laterally to the flanks (Grey Turner's sign), down to the inguinal ligaments (Fox's sign), or via the lesser omentum and 'bare area' of the liver to the falciform ligament and the skin around the umbilicus (Cullen's sign), manifesting as diffuse bruising.

Peripancreatic acute fluid collections frequently resolve spontaneously as the inflammation in the pancreas subsides. Accumulation of fluid anterior to the pancreas, just beneath the posterior wall of the lesser sac, can result in a 'pseudocyst' containing amylase-rich fluid and debris but lacking a true epithelial lining (Banks et al 2013). A large pseudocyst bulges forwards into the lesser sac in contact with the posterior wall of the stomach, lesser omentum and gastrosplenic ligament. If it does not resolve spontaneously, the pseudocyst can be drained internally through the posterior wall of the stomach by endoscopic or open surgery, or externally by means of a percutaneous drain inserted under radiological guidance.

Sustained injury to the pancreas can cause permanent damage, leading to parenchymal fibrosis and calcification within the parenchyma and the pancreatic ducts. This often causes chronic pain, diabetes mellitus and pancreatic exocrine insufficiency with steatorrhoea and malnutrition. Fibrosis can involve the bile duct in the head of the pancreas, leading to obstructive jaundice, and/or the splenic vein, causing splenic vein thrombosis and 'left-sided' portal hypertension.

The vast majority of pancreatic tumours are ductal adenocarcinomas derived from exocrine ductal epithelium. Up to two-thirds of such tumours are located in the head of the pancreas. Tumours in the head of the pancreas often obstruct the pancreatic segment of the bile duct early in the course of disease, leading to obstructive jaundice. The appearance of a dilated common bile duct and main pancreatic duct on cross-sectional imaging (the 'double duct' sign) is a characteristic sign in tumours of the pancreatic head. Advanced tumours may invade the adjacent duodenum, causing gastric outlet obstruction and/or the portal or superior mesenteric veins posterior to the neck of the pancreas. A tumour in the body or tail of the gland often grows to a large size before presenting as an abdominal mass or with back pain and weight loss (Yeo et al 2002). Tumours of the uncinate process can cause early involvement of the superior mesenteric vessels and duodenum (O'Sullivan et al 2009). Peripancreatic and pre-aortic lymph nodes may be involved, together with hepatic nodes, in pancreatic head tumours, and splenic hilar nodes may be involved in pancreatic tail tumours.

Surgery for pancreatic head tumours usually involves combined resection of the pancreatic head and duodenum (pancreaticoduodenectomy) because of their shared blood supply. The resectability of the tumour depends on the presence and extent of major vascular involvement, particularly the portal vein and superior mesenteric vessels. An accessory or replaced right hepatic artery (p. 1166) running cranially on the posterior surface of the pancreatic head and behind the supraduodenal segment of the bile duct is at risk of injury during such procedures (Turrini et al 2010). Recognition of this variation and preservation of the artery are essential to maintain the arterial supply to the right lobe of the liver. In distal or total pancreatectomy, spleen-preserving resections are generally undertaken if the underlying pathology does not involve the splenic vein lying in the groove on the posterior surface of the gland, taking care to control the multiple small pancreatic venous tributaries. In duodenum-preserving resection of the pancreatic head, which is sometimes undertaken for chronic pancreatitis and benign pancreatic tumours, it is important to preserve an adequate duodenal blood supply from the pancreaticoduodenal arterial arcades (Kimura and Nagai 1995). Stenosis of the coeliac artery is common in elderly individuals with atherosclerosis. Here, the arterial blood supply to the liver is provided by the superior mesenteric artery through hypertrophied arteries of the anterior and posterior pancreaticoduodenal arcades. Care should be taken to avoid ligation of these branches during pancreatic surgery since this could cause hepatic ischemia (**Fig. 69.11**).

Fig. 69.11 An intraoperative photograph taken during pancreaticoduodenectomy. This patient had coeliac artery stenosis and consequently hypertrophied pancreaticoduodenal arcade arteries (A) provided the arterial supply to the liver. The pancreatic head and duodenum were resected, preserving the arcade. Note the cut stump at the proximal body of the pancreas (B) and the exposed portal/superior mesenteric vein (C). The arcade communicates with the hepatic artery (E) to supply the liver (D). (Courtesy of Mohamed Rela, Global Hospitals, Chennai, India.)

The pancreas is usually transplanted along with the kidney (this is termed a simultaneous pancreas–kidney transplant) in patients with end-stage renal disease caused by diabetes mellitus. In selected patients, the pancreas alone can be transplanted to treat poor glycaemic control and hypoglycaemia unawareness (White et al 2009).

Donor pancreas is recovered from deceased donors, usually as part of a multi-organ recovery procedure. The entire pancreas, along with the duodenal C loop (attached to the head of pancreas) and spleen (closely related to the tail of the pancreas), is recovered to prevent damage to the vascular arcades in the pancreaticoduodenal groove and minimize handling of the pancreas during organ recovery (Abu-Elmagd et al 2000). The arterial supply for the pancreas graft is based on the splenic artery and the superior mesenteric artery, both of which should be carefully preserved during the donor operation. Both arteries are reconstructed to a Y-shaped deceased donor iliac artery graft to enable a single arterial anastomosis in the recipient. The venous drainage is through the portal vein. The pancreas graft is implanted heterotopically in the pelvis (not in its normal anatomical position, i.e. the upper retroperitoneum). The reconstructed artery is anastomosed to the external iliac artery. The portal vein of the graft can be anastomosed to either the iliac vein or the superior mesenteric vein of the recipient. Though portal and systemic venous drainage each has its own supporters, there is no evidence regarding the superiority of one technique versus the other (Bazerbachi et al 2012). Drainage of the pancreas exocrine secretions is into the ileum through a duodeno-ileal anastomosis or into the recipient urinary bladder through a duodeno-cystic anastomosis. Complications of the procedure are primarily a result of the damage the delicate pancreas sustains during the organ recovery, storage and transplantation process. Complications include graft pancreatitis, graft necrosis leading to bleeding, thrombosis of vascular anastomoses and sepsis (Delis et al 2004, Troppmann 2010). However, in the majority of patients, pancreas transplantation provides the potential for an insulin-free life and retards the development of diabetes-related complications (Gruessner et al 2012).

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